Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/05/2012
PI Name:	Mollicone, Daniel Ph.D.		
Project Title:	Android App Incorporating the PVT to Deliver Individualized Fatigue Risk Management in Commercial Trucking		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	09/01/2011	**End Date:**	02/29/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NBPF00007		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overarching objective of this project is to achieve an Android App that incorporates the Psychomotor Vigilance Task (PVT) to deliver Individualized Fatigue Risk Management in 24-7 occupations that involve fatigue stressors (e.g., extended duty hours, chronic sleep restriction, night work, and jet lag). This project has the following 3 specific aims: (1) Develop PVT feedback algorithm for use with truck driver populations; (2) Design smartphone fatigue management user interface with vigilance feedback and sleep entry; and (3) Develop prototype Android App with PVT and fatigue management interface. During this project we met all of the objectives. We completed an Android App with PVT and fatigue management interface. This app enables users to perform the PVT, set a wake-up alarm, input and track sleep history, and receive feedback from PVTs taken. The deliverables achieved in this project provide a critical element needed to complete the next stage of our research and development—the Android App with PVT and fatigue management interface to complete data acquisition on commercial truck drivers during long-haul trucking operations.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overarching goal of this project is to achieve a smartphone-based fatigue management tool app that incorporates the PVT. The deliverable achieved in this project will address a critical need for cost-effective tools for the serial assessment of neurobehavioral performance in operational environments using ubiquitous mobile devices. Such tools are especially needed in occupations where human performance has precise operational constraints and important safety implications, such as space exploration, military operations, air travel, and emergency health care.
Task Progress:	During this reporting period we achieved the following objectives/tasks: (1) Developed PVT feedback algorithm for use with truck driver populations; (2) Designed smartphone fatigue management user interface with vigilance feedback and sleep entry; and (3) Developed prototype Android App with PVT test and fatigue management interface.
Bibliography Type:	Description: (Last Updated: 02/23/2015)